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Solution

We can take air molecules in the tire as an ideal gas. The number of molecules and the volume of tire remain constant. So the air molecules at 27°C satisfies the ideal gas equation $P_1V_1 = NkT_1$ and at 39°C it satisfies $P_2V_2 = NkT_2$

But we know

$$V_1 = V_2 = V$$

$$\frac{P_1V}{P_2V} = \frac{NkT_1}{NkT_2}$$

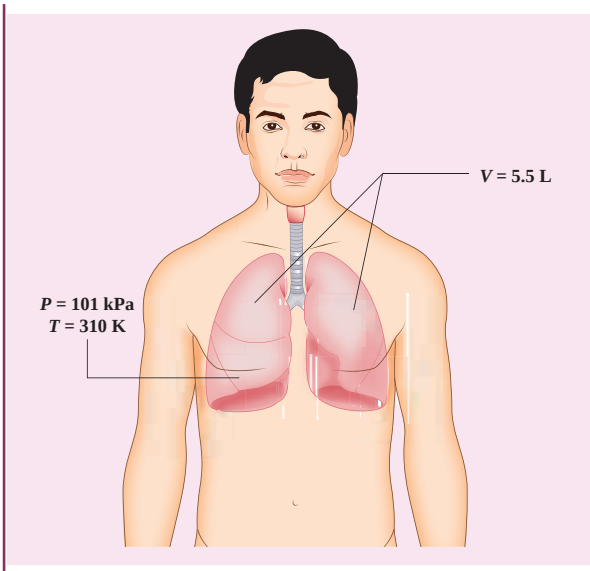
$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$P_2 = \frac{T_2}{T_1}P_1$$

$$P_2 = \frac{312K}{300K} \quad 240 \quad 10^3 Pa = 249.6\ kPa$$

EXAMPLE 8.3

When a person breaths, his lungs can hold up to 5.5 Litre of air at body temperature 37°C and atmospheric pressure (1 atm =101 kPa). This Air contains 21% oxygen. Calculate the number of oxygen molecules in the lungs.



Solution

We can treat the air inside the lungs as an ideal gas. To find the number of molecules, we can use the ideal gas law.

$$PV = NkT$$

Here volume is given in the Litre. 1 Litre is volume occupied by a cube of side 10 cm.
$1 Litre = 10cm \times 10cm \times 10cm = 10^{-3}\ m^3$

$$N = \frac{PV}{kT} = \frac{1.01\times10^5\ Pa \times 5.5\times10^{-3}\ m^3}{1.38\times10^{-23}\ JK^{-1}\times310K}$$

$= 1.29\times10^{23}$ Molecules

Only 21% of N are oxygen. The total number of oxygen molecules

$$= 1.29 \times 10^{23} \times \frac{21}{100}$$

Number of oxygen molecules
$= 2.7 \times 10^{22}$ molecules

EXAMPLE 8.4

Calculate the volume of one mole of any gas at STP and at room temperature (300K) with the same pressure 1 atm.